Data Ingestion and Cleaning Documentation

1. Ingestion Setup

Because raw files were stored in GitHub via Git LFS, they were pulled directly into memory using requests and io.BytesIO:

- Wages (2025 OEWS): Read the "All Areas" tab (statewide + 13 LWDAs).
- Demographics (Maryland Planning / Census): Read the "Regional_Share" tab, skipping two header banner rows (header=2).

2. Wage Data Cleaning

- Removed hourly metrics: Dropped all columns starting with "Hourly" to focus only on annual pay.
- Standardized column headers: Trimmed whitespace, converted to lowercase, and replaced consecutive spaces with single underscores.
- Handled missing data: Used a targeted drop only if both mean and median annual wages were missing.
    - Note: The 96 missing employment counts and 30 missing experience values reflect standard state data suppression for tiny sample sizes to protect privacy. Core median wages have zero missing values.
- Occupation hierarchy flags: Added boolean flags (is_total, is_major_group, is_detailed) to distinguish overall totals (00-0000) and 2-digit industry groups (0000) from 6-digit jobs to prevent double-counting in regressions and t-tests.
- Type casting: Converted employment counts and annual wage metrics to numeric (float/int).

3. Demographics Data Cleaning and Feature Engineering

- Filtered population rows: Retained only "Jurisdiction," "Sex," "Race," and "2025" counts, keeping rows where Sex == 'Total' to exclude gender-split subsets.
- LWDA geographic mapping: Mapped Maryland's 24 jurisdictions into the 13 official LWDAs (e.g., grouping Allegany, Garrett, and Washington into Western Maryland).
- Pivoted race categories: Filtered for "Total Population" and "Total Racial and Ethnic Minority Population **" and pivoted them into separate columns summed by LWDA.
- Engineered demographics metric:

$$minority_pct = \left(\frac{minority_population}{total_population}\right) \times 100$$

4. Merging for Modeling

- Joined df_clean and df_lwda with area_name.
- Confirmed all 14 geographic areas (13 LWDAs + Statewide) merged cleanly with 0 missing demographic values.

Reproducibility Steps:

1. Run cells sequentially in 01_Ingestion.ipynb in Colab or Jupyter.
2. Ensure libraries are imported and installed.
3. Execute Part 1a to produce MD_OEWS_Cleaned.csv.
4. Execute Part 1b to produce MD_LWDA_Demographics_Summary.csv.
5. Execute Part 1c to produce MD_Wage_Demographics_Master.csv for modeling.